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VOICE OF VILLAGE: FUND TRANSPARENCY AND ISSUE REPORTING SYSTEM

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ABSTRACT

Rural governance in India faces persistent challenges related to fund mismanagement, inefficient Public Distribution System (PDS) operations, absence of citizen grievance mechanisms, and inadequate digital infrastructure. Existing systems are fragmented, non-integrated, and lack real-time monitoring capabilities. This paper presents "Voice of Village", a comprehensive full-stack web application designed to bridge these gaps by integrating fund transparency, issue reporting, water and electricity service requests, house construction approvals, PDS ration management, and community donation management into a single unified platform. The system employs a role-based access control (RBAC) model supporting four user types: Villager, Sarpanch, Admin, and Government Official. The backend is developed using Node.js with Express.js, MySQL as the relational database hosted on Railway cloud infrastructure, and JWT-based authentication. The frontend is built using React.js with interactive data visualization through Recharts. The proposed system enables real-time transparency, improves citizen-authority communication, and promotes accountable digital governance in rural communities.

Keywords: Rural Governance; Fund Transparency; Public Distribution System; Issue Reporting; Role-Based Access Control; React.Js; Node.Js; JWT Authentication; Real-Time Monitoring; Digital Governance.

I. INTRODUCTION

In recent years, digital technologies have significantly transformed governance and public service systems worldwide. Governments and administrative bodies are increasingly adopting digital platforms to improve efficiency, transparency, and accessibility. However, despite these advancements, many rural areas in developing countries, particularly in India, continue to face severe challenges in the effective implementation of governance systems.

One of the most critical concerns in rural governance is the lack of visibility into how government funds are allocated and utilized. Villagers often do not have clear information about budgets, expenses, or development activities occurring in their communities. This opacity creates opportunities for mismanagement, corruption, and reduces citizens' trust in local authorities. Without proper access to financial information, citizens are unable to actively participate in governance or hold officials accountable.

The Public Distribution System (PDS) plays a vital role in providing essential commodities such as rice, wheat, sugar, and other necessities to economically weaker sections. However, traditional PDS systems suffer from manual record-keeping, delays in distribution, lack of proper verification, and significant chances of fraud or duplication of beneficiary data. The absence of a real-time digital tracking system further exacerbates these issues.

Beyond fund transparency and ration distribution, the absence of a structured platform for reporting local infrastructure problems further limits effective governance. Villagers frequently encounter issues related to roads, sanitation, water supply, and electricity. Without a proper reporting mechanism, these issues go unaddressed, leading to delays and dissatisfaction among citizens.

1.1 Motivation

The primary motivation for this work stems from the observation that rural villages in Telangana and across India continue to operate governance processes through manual or semi-digital means. Lack of accountability

in fund utilization, poor grievance redressal systems, and limited digital access for rural citizens inspired the development of a platform that could bring transparency, convenience, and efficiency to village-level administration.

1.2 Objectives

The key objectives of this system are:

- To provide a transparent fund management module accessible to all villagers.
- To digitize the Public Distribution System for efficient ration tracking and distribution.
- To enable villagers to report and track local infrastructure and service issues.
- To streamline water, electricity, and house construction service requests.
- To provide an interactive reporting dashboard with data visualizations for all stakeholders.
- To implement secure role-based access control for Villager, Sarpanch, Admin, and Government Official roles.

1.3 Scope of the System

The scope of this project covers village-level governance operations including fund transparency, PDS management, grievance reporting, utility request management, and community donation tracking. The system is designed for deployment in village panchayats and gram panchayats across Telangana, India, with potential for broader geographic adaptation.

II. LITERATURE SURVEY

Several research works and existing systems have attempted to address isolated aspects of rural governance. This section reviews the most relevant literature and identifies the gaps this system addresses.

2.1 Fund Transparency Systems

Setiawan and Fitriani (2018) examined village fund management transparency in Indonesia and concluded that public accessibility of financial data significantly reduces corruption and improves community trust [1]. Singh and Kumar (2021) further analyzed accountability and transparency in village fund management and found that integrated digital platforms are superior to manual reporting in ensuring fund utilization visibility [2]. Junaidi and Adnan (2023) studied transparency and accountability of village fund management and emphasized the need for unified platforms combining financial tracking with citizen communication tools [5].

2.2 Public Distribution System (PDS) Digitization

Mhetre et al. (2023) proposed an E-Ration Distribution System based on QR codes, demonstrating that digital verification significantly reduces ghost beneficiaries and fraud in ration distribution [3]. Their system, however, focused solely on ration collection verification without integrating fund transparency or issue reporting features.

2.3 ICT for Rural Development

Rahman and Islam (2021) conducted a comprehensive study on ICT applications for rural development and highlighted the challenges of limited digital literacy, connectivity issues, and absence of user-friendly interfaces as primary barriers to technology adoption in rural areas [4]. They recommended citizen-centric designs with minimal navigation complexity.

2.4 Research Gap

A review of existing systems reveals that most solutions address only one or two aspects of rural governance in isolation. No existing platform combines fund transparency, PDS management, multi-category issue reporting, utility service requests, house construction approvals, donations, and interactive report dashboards into a single integrated system. The proposed Voice of Village system addresses this gap comprehensively.

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III. SYSTEM ANALYSIS

3.1 Existing System Limitations

The analysis of existing rural governance systems highlights several limitations:

- Fund management records are maintained in manual or semi-digital formats, with no public access portal.
- PDS operations rely on manual beneficiary registers, making fraud detection difficult.
- No structured grievance reporting platform exists for villagers to submit and track complaints.
- Utility service requests (water, electricity) require physical office visits, causing delays.
- House construction approvals involve multi-level bureaucratic processes with no digital tracking.
- Existing systems lack interactive dashboards and real-time reporting capabilities.

3.2 Proposed System Advantages

The proposed Voice of Village system overcomes these limitations by providing:

- A centralized platform integrating all governance services.
- Real-time fund transparency with public-facing dashboards.
- Digital PDS management with beneficiary verification.
- Structured issue reporting with status tracking and commenting.
- Online submission and tracking of water, electricity, and house requests.
- Interactive reports with pie charts, bar graphs, and trend lines.
- Secure JWT-based authentication with four distinct user roles.

3.3 System Architecture

The system follows a three-tier client-server architecture. The presentation layer is a React.js single-page application. The business logic layer is a RESTful API server built with Node.js and Express.js. The data layer consists of a MySQL relational database hosted on Railway cloud infrastructure, ensuring availability from any location.

Table 1: System Architecture Layers

Layer	Technology	Responsibility
Presentation	React.js, Recharts	UI, data visualization
Business Logic	Node.js, Express.js	REST APIs, auth, validation
Data	MySQL on Railway	Data storage and retrieval

IV. METHODOLOGY

The system was developed following an incremental development methodology, where each module was designed, developed, and tested independently before integration. The following subsections describe the key design and implementation decisions.

4.1 Technology Stack

Table 2: Technology Stack

Component	Technology	Purpose
Frontend	React.js	Single-page application UI
Charts	Recharts	Interactive data visualization
API Calls	Axios	Frontend-backend communication
Backend	Node.js + Express.js	RESTful API server
Authentication	JWT (jsonwebtoken)	Stateless authentication
Password Security	bcryptjs	Password hashing
File Uploads	Multer	Issue image attachments
Database	MySQL	Relational data storage
DB Hosting	Railway	Cloud MySQL hosting
Payment	Razorpay (test mode)	Donation payment gateway

4.2 Database Design

The database consists of the following core tables:

- users — stores user credentials, roles, and profile information.
- issues — stores citizen-reported problems with category, status, and image references.
- issue_comments — stores threaded comments on issues for authority-citizen communication.
- water_requests — stores water supply problem requests with type and priority.
- electricity_requests — stores electricity complaint and connection requests.
- house_requests — stores building plan approval and occupancy certificate applications.
- pds_records — tracks monthly ration collection status per beneficiary.
- donations — records donation transactions with Razorpay payment verification.

All tables use foreign key relationships to enforce referential integrity. The users table is the central entity referenced by all other tables via the submitted_by or user_id columns.

4.3 Authentication and Role-Based Access Control

The system implements JWT (JSON Web Token) based stateless authentication. Upon successful login, the server generates a signed token containing the user's ID and role. This token is validated on every protected API request through an authentication middleware. The four user roles and their permissions are as follows:

Table 3: Role-Based Access Control Matrix

Role	Permissions
Villager	Submit issues, raise service requests, view fund data, track PDS, donate
Sarpanch	All Villager permissions + update issue status, manage requests
Admin	Full system access, manage all users and modules
Govt. Official	View transparency reports, fund summaries, and analytics dashboards

4.4 Module-wise Implementation

Each module follows the MVC (Model-View-Controller) pattern with dedicated route, controller, and database interaction files.

4.4.1 Issue Reporting Module

Citizens can submit issues categorized as road, water, electricity, sanitation, or other. Each issue includes a title, description, optional image upload (via Multer), and is assigned a default status of pending. Sarpanch and Admin users can update the status to in_progress or resolved. All users can add comments to any issue for collaborative resolution tracking.

4.4.2 Water and Electricity Request Modules

Villagers can raise formal requests for water supply problems (no supply, low pressure, contamination, pipe leakage, new connection) or electricity problems (power cuts, transformer failure, street light faults, meter complaints, new connection). Each request captures the affected area, number of households impacted, duration of the problem, priority level, and optional supporting photograph.

4.4.3 House Request Module

The house request module enables villagers to apply for building plan approvals, NOC from the gram panchayat, commencement certificates, completion certificates, occupancy certificates, and property tax registration. Applicants submit plot details including survey number, plot size, sale deed, house plan blueprints, and construction type (new house, extension, or renovation).

4.4.4 PDS (Public Distribution System) Module

The PDS module digitizes ration distribution tracking for the village. Beneficiaries are categorized by card type (AAY, PHH, APL) and monthly ration entitlements are calculated accordingly. Monthly collection records are maintained per beneficiary, tracking items including rice, wheat, sugar, edible oil, and kerosene. The system maintains a pds_records table to track whether each registered beneficiary collected their monthly entitlement.

4.4.5 Fund Transparency and Donations Module

The fund transparency module displays village fund allocations, expenditures, and remaining balances in a publicly accessible dashboard. The donations module allows community members to contribute funds for village development through Razorpay payment gateway integration (test mode). All donations are recorded in the database with payment verification through Razorpay's order-verify API flow.

4.4.6 Reports and Analytics Module

The reports module aggregates data across all modules and presents interactive visualizations. Report types include issue resolution rates (pie charts), monthly issue trends (bar graphs), water and electricity request resolution percentages, house request approval statistics, PDS collection rates, and donation trend line charts. Reports support filtering by weekly, monthly, yearly, and all-time periods.

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4.5 API Design

All backend endpoints follow RESTful conventions. The primary API groups are:

Table 4: Key API Endpoints

Method	Endpoint	Description
POST	/api/auth/register	Register a new user
POST	/api/auth/login	Authenticate and receive JWT
POST	/api/issues	Submit a new issue
GET	/api/issues	Fetch all issues
PUT	/api/issues/:id/status	Update issue status
POST	/api/issues/:id/comments	Add comment to issue
POST	/api/water-requests	Raise water request
POST	/api/electricity-requests	Raise electricity request
POST	/api/house-requests	Submit house construction request
GET	/api/reports/issues	Issue analytics report
GET	/api/reports/pds	PDS collection report
POST	/api/donations/create-order	Initiate Razorpay donation

V. RESULTS AND DISCUSSION

The Voice of Village system was tested through functional testing of all API endpoints using Thunder Client, database verification using MySQL Workbench, and end-to-end UI testing across all four user roles.

5.1 Functional Testing Results

All API endpoints were validated for correct responses, proper JWT authentication enforcement, and accurate database interactions. The table below summarizes the functional testing outcomes:

Table 5: Functional Testing Results

Test Case	Result	Observation
User Registration	Pass	Users created with hashed passwords
JWT Login	Pass	Token generated with role and user ID
Issue Submission	Pass	Issue saved with pending status

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Issue Status Update	Pass	Only Sarpanch/Admin can update
Add Comment to Issue	Pass	Threaded comments functional
Water Request Submit	Pass	Saved with correct priority
Electricity Request	Pass	All request types functional
House Request Submit	Pass	Document references stored
PDS Record Tracking	Pass	Monthly collection marked correctly
Reports - Issues	Pass	Percentage and trend data accurate
Donation Flow	Pass	Razorpay test payment successful
Role Access Control	Pass	Unauthorized access returns 403

5.2 System Performance

The system demonstrated reliable performance across all tested scenarios. API response times for standard GET requests averaged below 200ms with the Railway-hosted MySQL database. Authentication overhead through JWT verification added negligible latency (under 5ms per request). Image uploads through Multer for issue reports were handled efficiently for files up to 5MB.

5.3 User Interface Screenshots

The following figures illustrate the key interfaces of the Voice of Village system.

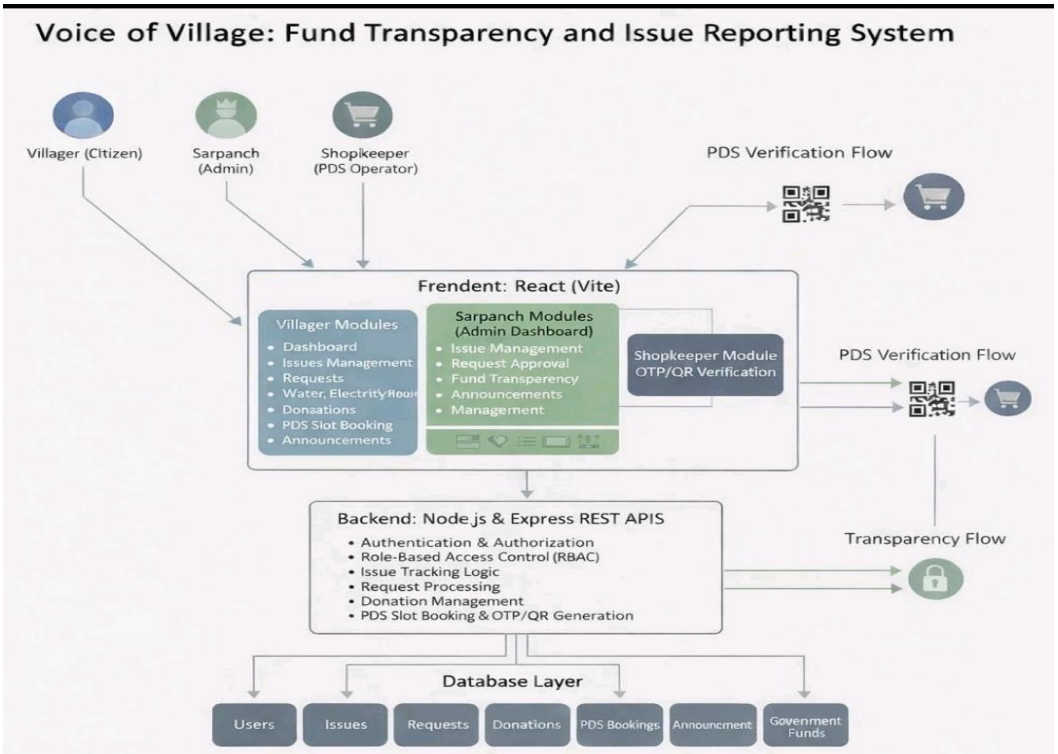

Figure 1: Overall System Architecture

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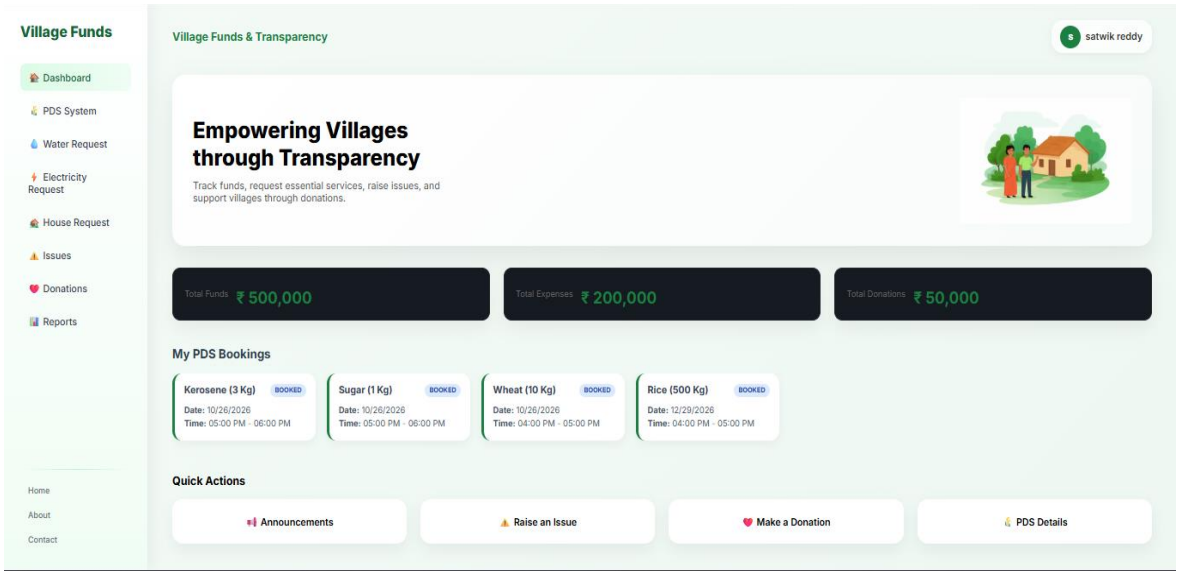

Figure 2: Citizen Dashboard with Fund Transparency Panel

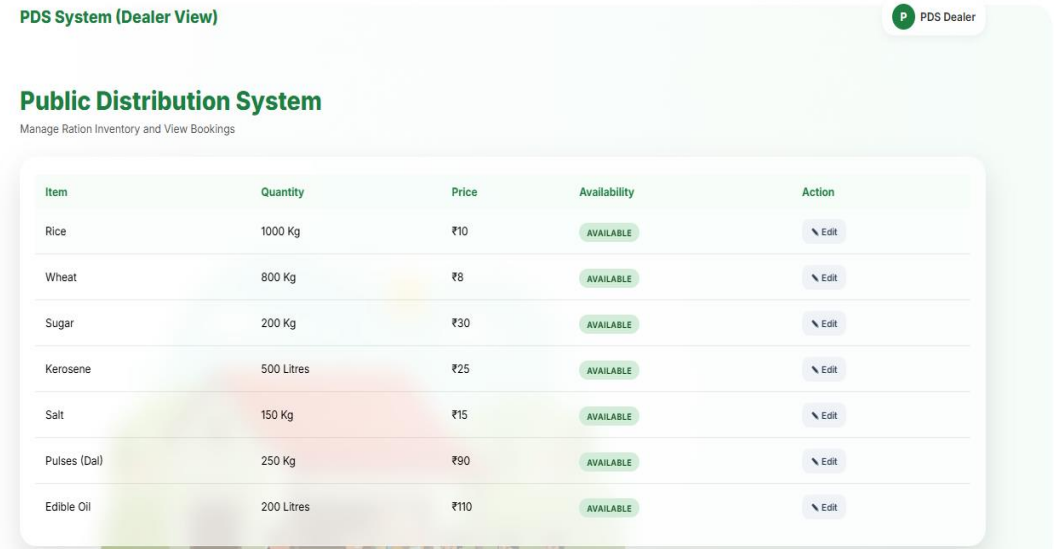

Figure 3.1: PDS Ration Distribution Tracking Interface

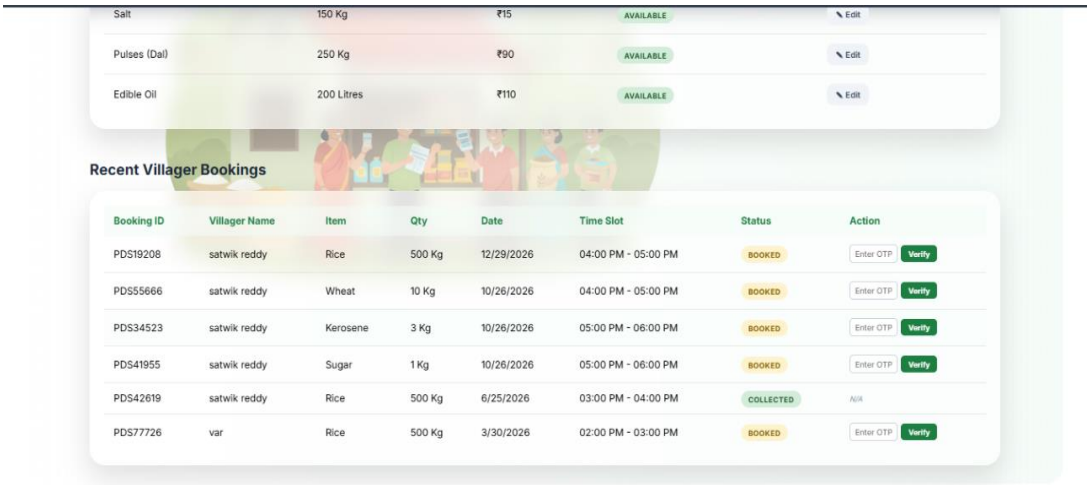

Figure 3.2: PDS Ration Distribution Tracking Interface

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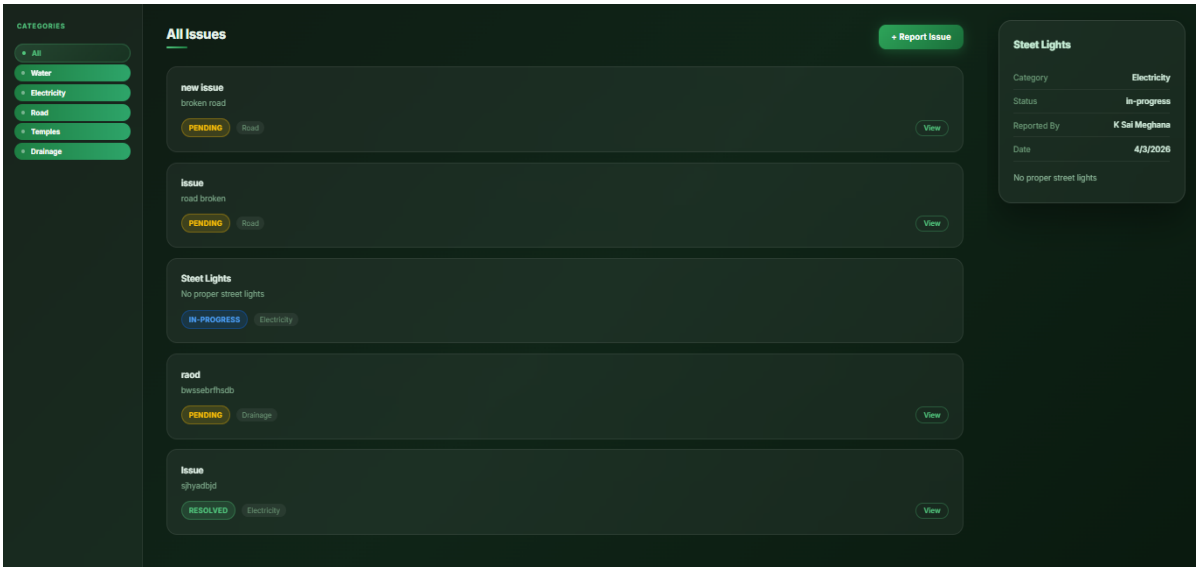

Figure 4: Issue Submission and Tracking Interface

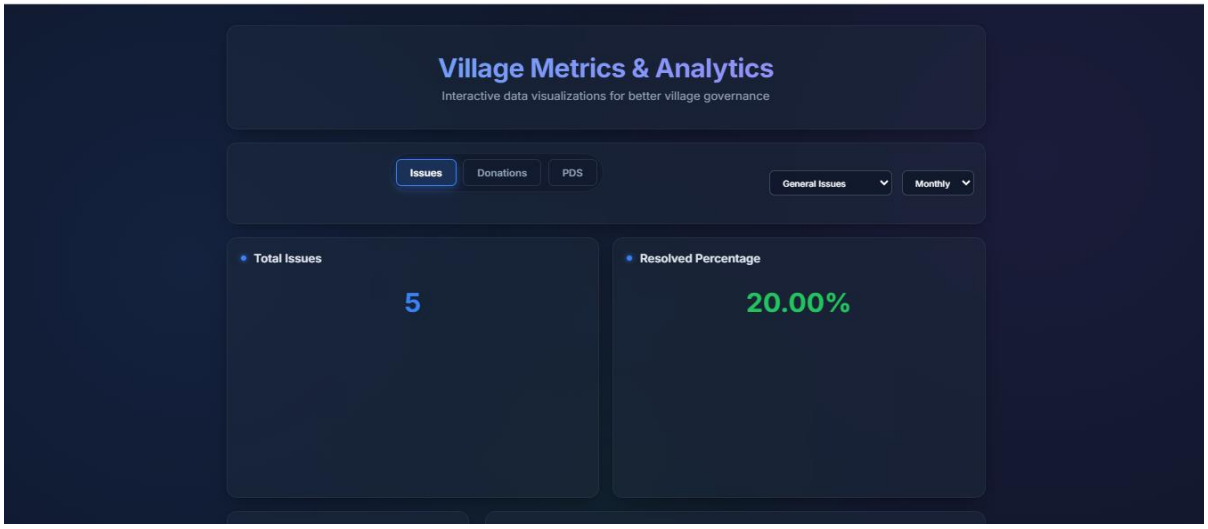

Figure 5.1: Interactive Reports Dashboard with Charts

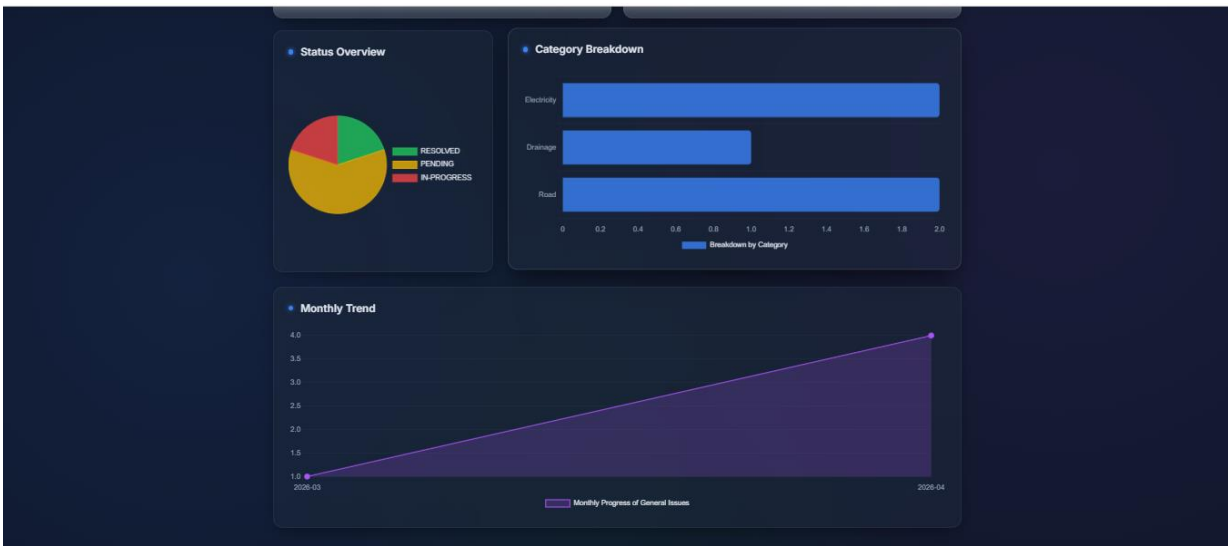

Figure 5.2: Interactive Reports Dashboard with Charts

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5.4 Comparison with Existing Systems

Table 6: Comparison of Voice of Village with Existing Systems

Feature	Voice of Village	Fund Mgmt. Systems	QR PDS [3]	ICT Rural [4]
Fund Transparency	Yes	Yes	No	Partial
Issue Reporting	Yes	No	No	No
PDS Management	Yes	No	Yes	No
Utility Requests	Yes	No	No	No
House Approvals	Yes	No	No	No
Role-based Access	Yes	Partial	No	Partial
Interactive Reports	Yes	No	No	No
Payment Integration	Yes	No	No	No
Cloud Database	Yes	No	No	No

VI. CONCLUSION

This paper presented "Voice of Village", a comprehensive full-stack web application designed to modernize rural governance through digital integration of fund transparency, issue reporting, utility service requests, house construction approvals, PDS management, and community donations. The system successfully addresses the critical limitations of existing rural governance systems by providing a unified, role-aware, and real-time platform accessible to all village stakeholders.

The proposed system demonstrates that integrating multiple governance services into a single platform with interactive reporting significantly improves transparency, citizen participation, and administrative efficiency. The use of JWT-based role-based access control ensures data security and appropriate access for each user type.

6.1 Future Work

Future enhancements planned for the system include:

- Mobile application development using React Native for broader rural accessibility.
- Aadhaar-based biometric verification for PDS beneficiary authentication.
- Multilingual support including Telugu and Hindi for improved rural usability.
- Push notification system for issue status updates and PDS collection reminders.
- AI-based analytics for predictive fund utilization and issue trend analysis.
- Integration with government APIs such as PM-Kisan and MGNREGS for direct data linkage.

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